

EtherCAT Master App

EtherCAT Master for ctrlX OS Version 4
Version-4.6.0

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DC-AE/PAG-SW (MiNi/PiaSt)

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1 About this documentation

Editions of this documentation

Edition	Release date	Note
01	2025 – 11	EtherCAT Master App, Version 4.4.0
02	2026 – 03	EtherCAT Master App, Version 4.6.0

2 Important information

2.1 Product information

For the latest information on our products, go to ➔ <https://www.boschrexroth.com>.

For current security advisories, please refer to ➔ <https://psirt.bosch.com/security-advisories/>

3 Version 4.6.0

3.1 New functions

3.1.1 Feature 946501: Extending ctrlX I/O Engineering with an FoE overview page

Description

A dialog with an FoE overview has been implemented on the EtherCAT master node, from which the active FoE transmissions can be diagnosed.

3.1.2 Feature 1047377: Improved handling of disabled/optional slaves in ctrlX I/O Engineering and in the PLC

Description

To simplify the handling of deactivated /optional slaves, the following functionalities have been added in I/O Engineering:

- Considering disabled slaves during scanning (display, comparison).
- Station alias can be set directly when scanning the field bus.

In addition, function blocks were implemented in the PLC to handle address assignment in EtherCAT, which are used to control and diagnose the functionality. The following function blocks have been newly implemented:

- IL_ECATCHGetConfiguredStationAlias
- IL_ECATCHSetConfiguredStationAlias
- IL_ECATCHGetExplicitDeviceIdentification

3.1.3 Feature 1058291: Test license available

Description

If no EtherCAT Master license is available, a 10-day test license can now be used in the app settings.

3.1.4 Feature 1117197: EoE ping functionality in the EtherCAT web interface

Description

The ctrlX web interface for a device that supports EoE now includes a ping button to check the basic IP network reachability and latency of the device in relation to ctrlX control.

3.1.5 Feature 1119160: CANopen gateway support and improvements in the HexViewer in ctrlX I/O Engineering

Description

HexViewer improvements (AoE, IO-Link, VoE):

- Support for loading and saving files
- Extended clipboard support

CANopen:

- For devices that support the CANopen profile, an additional page for reading and writing objects to CANopen nodes is displayed.
- The supported profile is the "CANopen Master profile (Module Profile Number 5100)" of the EtherCAT Modular Device Profile.

3.2 Resolved defects and modifications

3.2.1 Bug 1049216: The enumeration text in CoE tables was corrected

Description

The CoE slave page view does not display any enumeration values when using the online object directory of the EtherCAT slaves.

Troubleshooting

The enumeration values are now displayed and can be selected for writable objects using a pull-down menu.

3.2.2 Bug 1120851: EtherCAT Working Counter: The status of the WkcValidIn/WkcValidOut bits is delayed for the error response of the cyclic data

Description

In the event of Wkc errors in cyclic communication, the status of the WkcValidIn/WkcValidOut bits is changed one cycle later than the error response of the cyclic data.

Troubleshooting

The status change of the WkcValidIn/WkcValidOut bits takes place in the same cycle as the error response of the cyclic data.

3.3 Notes on use and known restrictions

3.3.1 Bug 368154: Problems in ctrlX I/O Engineering when scanning the EtherCAT switches CU1128 and EP9128_0021

Description

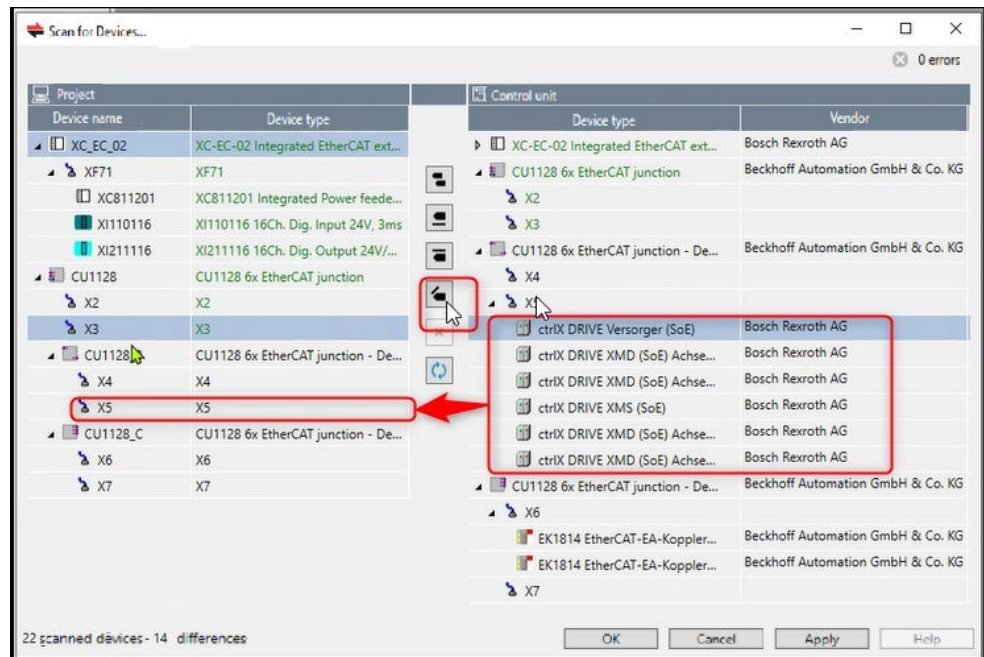
When scanning CU1128 / EP9128, an error occurs when adding the devices. Port X2 and X3 are added correctly, all other ports are not added to the project.

Workaround

After the error has occurred, you can scan again and add each branch individually to the project.

Port X4, X5, X6, X7 have to be added below the connector, as shown in the screenshot.

The devices connected to port X8 have to be added after the first CU1128 device (marked button in the following screenshot).



3.3.2 Bug 745765: Timestamps (Task.StartTime) in the DataLayer RT image do not work with bus shift applications

Description

If the "Bus Shift" mode is selected for Distributed Clocks, the time stamps in the DataLayer RT image are set to 0 as the mode is not supported for the time information at the current time.

Workaround

Use the DC mode "Auto" or "Link Layer Reference Clock".

3.3.3 Bug 1026963: EtherCAT web interface: CoE export cancellation still triggers download

Description

It is not possible to cancel a CoE export. Several exports can be started, which are then executed simultaneously.

3.3.4 Bug 1100086: EtherCAT web interface: FoE upload does not work

Description

When applying the FoE upload, an error message is displayed and the upload is canceled. This resulted in an error message.

3.3.5 Bug 1148941: EtherCAT web interface: Not all FoE file names can be selected

Description

The slave file name "debug/test filename.that" is not permitted, but it should be.

4 Version 4.4.0

4.1 New functions

4.1.1 Feature 454744: Support for predefined PDO configurations (VendorSpecific/AlternativeSmMapping)

Description

Toggling between different configurations defined in the ESI file is easily possible via „VendorSpecific/AlternativeSmMapping“.

4.1.2 Feature 706107: New dialogs in the EtherCAT web interface for VoE, AoE, FoE and EoE

Description

New diagnostic pages are provided for the web interface:

- VoE and FoE information for the EtherCAT Master.
- AoE, EoE and VoE information for EtherCAT slaves.

4.1.3 Feature 916819: ctrlX I/O Engineering – Defining EtherCAT addresses and deactivating slaves

Description

It is now possible to deactivate individual EtherCAT slaves in the project.

In addition, EtherCAT addresses can also be permanently assigned so that they are not automatically changed when deactivated or activated.

4.1.4 Feature 938778: EtherCAT Master app – Connection to the ctrlX CORE authorization system

Description

The EtherCAT Master app provides the View, Operate and Manage authorizations.

Depending on the active authorizations of a user, the available functions in the EtherCAT web interface are adjusted accordingly.

In ctrlX I/O Engineering, corresponding error messages are generated if authorizations are missing.

4.1.5 Feature 945150: Extended functionality of multiple EtherCAT master instances

Description

Multiple EtherCAT master instances can now use the same distributed clock (Distributed Clock) as a time base.

The web user interface has an improved start page dialog tile, which enables the status of each EtherCAT master instance to be displayed.

In addition, the tile now contains a new tab that provides information on the distributed clock.

4.1.6 Feature 948568: ctrlX I/O Engineering – SCI file support

Description

SCI files are directly visible after the import and are highlighted accordingly both in the device database and in the project.

4.1.7 Feature 948582: The handling of sync units has been improved

Description

New pages on "Sync Units" have been added to the web user interface.

4.1.8 Feature 970417: Optimizations in the ctrlX I/O Engineering user interface

Description

A number of changes have been made to the ctrlX I/O Engineering user interface to improve user-friendliness:

- A revision deviation between the ESI file of the project and the real device is displayed in the device scan dialog. Devices that do not match are highlighted in blue and are provided with an information symbol and a tooltip. This speeds up the detection of revision deviations.
- In large projects, it was previously difficult to find a slave from an overview page (e.g. if the slave statistics page is open and the user detects a problem with a slave and wants to jump directly to this slave). To facilitate such actions, a context menu has been added to all relevant overview pages.
- A context menu for switching the EtherCAT slave mode has been integrated into many dialogs. This speeds up slave management.
- The settings for the distributed clocks (DC) have been moved to a separate dialog to facilitate managing DC settings.

4.1.9 Feature 973718: ctrlX I/O Engineering – Calling up ctrlX DRIVE Engineering is possible via the context menu

Description

The context menu of a ctrlX DRIVE node has been extended by a command that calls up the engineering tool "ctrlX DRIVE Engineering" directly for the selected drive.

4.1.10 Feature 987508: ctrlX I/O Engineering – Evaluating error codes for AoE and IO-Link

Description

When using communication on the AoE or IO-Link side of a corresponding slave, the evaluation of error codes has been improved.

4.1.11 Feature 988671: Improved Memory Viewer for Data Layer RT

Description

The Data Layer RT Memory Viewer has been improved with regard to the display of variable lists and the editing of variables.

4.1.12 Feature 1010879: ctrlX I/O Engineering – Revised IO-Link support via IODD for IO-Link master Beckhoff EL6224

Description

The following changes have been made to the configuration of the IO-Link master using IODD:

- Bug fixes in relation to IO-Link device support.
- The naming of the variables was changed to be incompatible, which was necessary to make the variable names independent of the used port.

4.1.13 Feature 1054196: ctrlX CORE X7 – Support of 250µs cycle time

Description

The ctrlX CORE X7 control variant previously only supported a minimum cycle time of 500 µs.

This limit was lowered to 250 µs so that the shorter cycle time of 250 µs can now be selected.

4.1.14 Feature 1073705: ctrlX CORE X3 – Support of 1ms cycle time

Description

The ctrlX CORE X3 control variant previously only supported a minimum cycle time of 2 ms.

This limit has been lowered to 1 ms so that the shorter cycle time of 1 ms can now be selected.

4.2 Resolved defects and modifications

4.2.1 Bug 472250: I/O Engineering – Differences in scanning are difficult to detect

Description

Differences during scanning are displayed in black and are therefore difficult to detect. Furthermore, devices are displayed as compatible if the revision differs.

Bugfixing

Project differences are highlighted in red.

If the revision deviates, the device is displayed in blue with an info text.

4.2.2 Bug 708690: The Data Layer browser does not display any process data for S20-IOL-8

Description

The Data Layer browser does not display any process data for S20-IOL-8.

Bugfixing

Change the data type for the byte array from `types/plc/raw` to `types/data-layer/array-of-uint8`.

4.2.3 Bug 857455: Compiler error if the variable name is used more than once (e.g. reserved)

Description

If a variable name is assigned multiple times, an error message is generated for each variable. To correct this, each variable has to be renamed individually. Some configurations can have several hundred entries, so renaming the variables takes a lot of time.

Bugfixing

The errors are still reported. However, an automatic correction is provided. This allows duplicate variables to be renamed either for one slave or for all slaves on the EtherCAT Master.

4.2.4 Bug 911323: Uninformative error message during configuration download with incorrect cycle time

Description

During the configuration download with certain cycle times such as 2500 μ , an uninformative error message was reported. The behavior always occurred at cycle times that were a multiple of 250 μ , but not a multiple of 1 μ .

Bugfixing

The cycle time is checked before the download is carried out and a corresponding error is reported.

4.2.5 Bug 1001465: Error in updating the slave statistics values was fixed

Description

The page with the slave statistics did not update the displayed values in all cases.

Bugfixing

The recording of slave statistics is now triggered internally and the values are updated.

4.2.6 Bug 1004526: EtherCAT Master – Sporadic crash in case of parallel mailbox communication

Description

If the CoE or SoE mailbox communication to several EtherCAT devices takes place simultaneously, the EtherCAT Master app crashed sporadically.

Bugfixing

The cause of the sporadic crashes has been fixed.

4.2.7 Bug 1006871: CoE page – Objects with gaps in the sub-index cannot be expanded

Description

On the CoE side, objects with gaps in the subindex could not be expanded if only the online dictionary was used. For non-existent subindices, the error code "The object does not exist in the object dictionary" was returned instead of the correct code "Subindex does not exist". This resulted in problems as

some CoE indices could not be expanded if there were gaps in the subindex (e.g. 0x2C10, timing relevant, osci etc.). The system did not continue reading the next elements if an unexpected error occurred, resulting in an incomplete data display.

Bugfixing

The error handling has been updated. If an unexpected error occurs when reading sub-indexes, the system now only discards the incorrect sub-index and continues reading the other elements. This ensures that objects with gaps in the sub-index can be expanded and displayed correctly on the CoE page.

4.2.8 Bug 1022631: PDO entries with a size of 31 bytes were split up

Description

PDO entries can only be created in the slave with a maximum size of 31 bytes. If there is a larger entry in the ESI, the entry is split by the configuration tool. Previously, splitting was also carried out if the entry was exactly 31 bytes in size.

4.2.9 Bug 1054696: ctrlX I/O Engineering – "No converter available" is displayed for array variables

Description

In case of array variables (e.g. "ARRAY[] OF BYTE" in the variable browser), the error message "No converter available" is displayed.

Bugfixing

Missing converters have been added in the "Variables" section.

4.2.10 Bug 1064373: CoE overview – Reading of SubIndex not possible if length > 1024 bytes

Description

It was not possible to expand an object in the CoE overview if one of the sub-indices contained a data with a length of > 1024 bytes.

Bugfixing

The object can now be expanded and the value is displayed.

4.3 Notes on use and known restrictions

4.3.1 Bug 368154: Problems in ctrlX I/O Engineering when scanning the EtherCAT switches CU1128 and EP9128_0021

Description

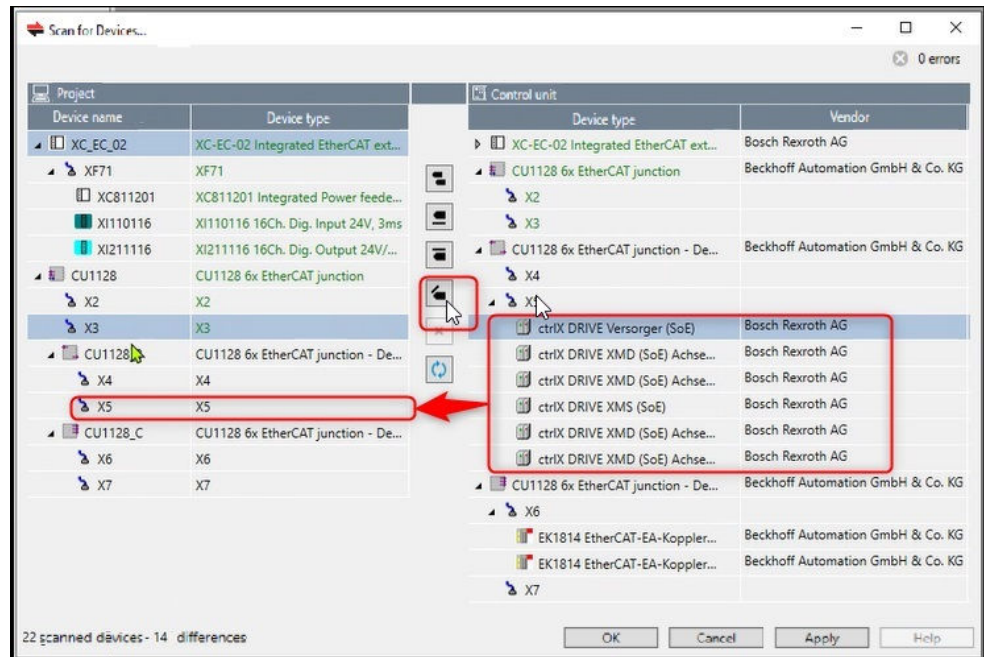
When scanning CU1128 / EP9128, an error occurs when adding the devices. Port X2 and X3 are added correctly, all other ports are not added to the project.

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4.3.3 Bug 1049216: The enumeration text in CoE tables was corrected

Description

The CoE slave page view does not display any enumeration values when using the online object directory of the EtherCAT slaves.

5 Service and support

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Preparing information

To be able to help you more quickly and efficiently, please have the following information ready:

- Detailed description of malfunction and circumstances
- Type plate specifications of the affected products, in particular type codes and serial numbers
- Your contact data (phone and fax number as well as your e-mail address)

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